



# RC Racers

TDP  
Feb. 2026

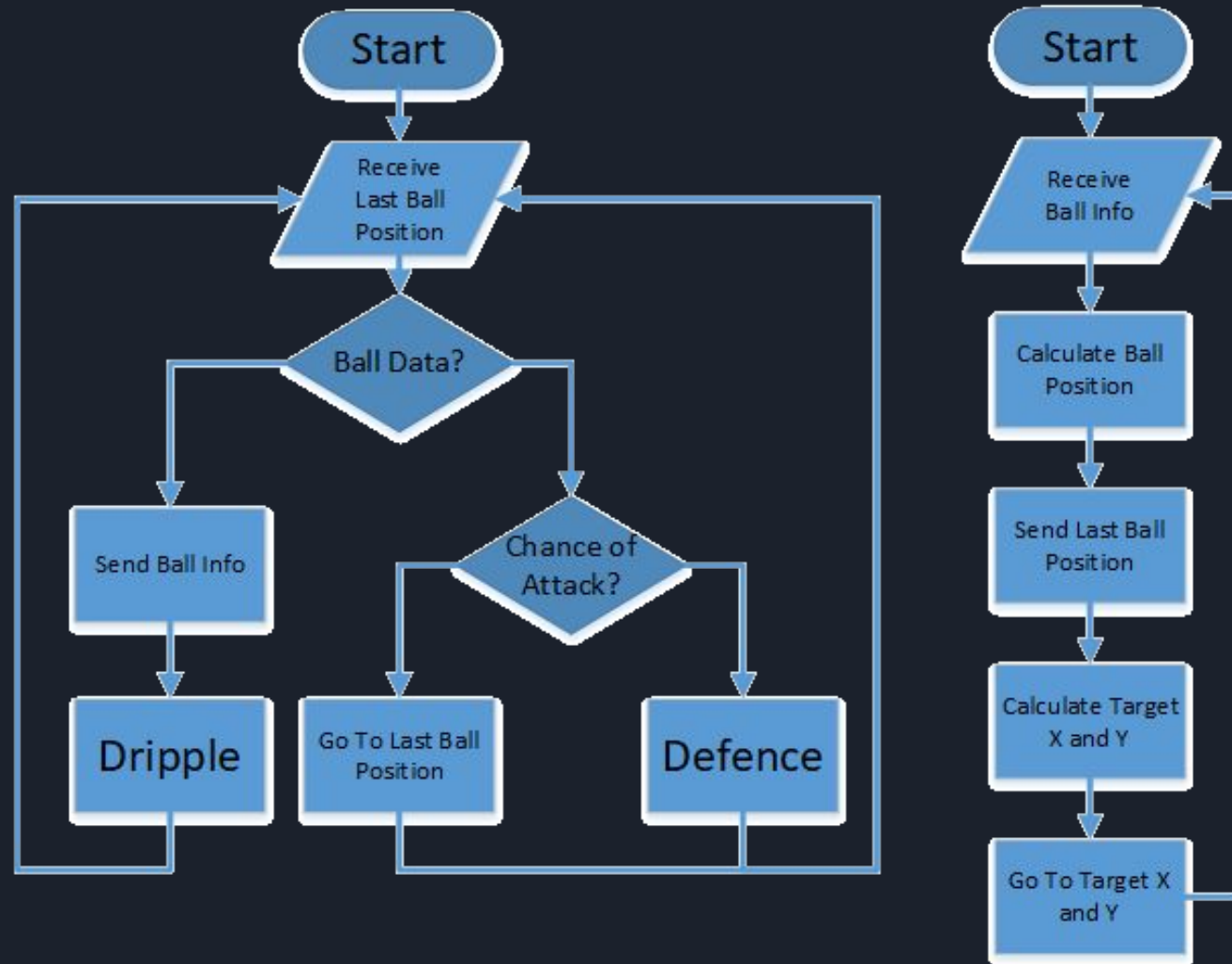
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# Summary

- Our code requires numpy library to run
- Our 1<sup>st</sup> Robot is the Goal-Keeper
- The other two are attackers and sometimes they do defense
- 2<sup>nd</sup> and 3<sup>rd</sup> Robot send ball and robot global data to 1<sup>st</sup> Robot
- 1<sup>st</sup> Robot calculates ball position and sends it to the team
- If there are 2 ball data a system of equations is solved using numpy
- If there is only 1/no ball data its position is estimated/predicted
- We utilized the rotation matrix a few times in our code

# Attackers & Goalkeeper



# Future Work

1. A more complex Algorithm to work better as a team
2. Having a dynamic Goal-Keeper instead of a fixed one
3. Detect Other team robots and track their position
4. A more advanced motion-planning
5. Integrating an effective AI model

Thank you!